

# Decision-Effective Conformal Ambiguity Sets for Neural PDE Control

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## Abstract

Neural operators can predict PDE fields accurately while remaining unreliable in the directions that determine a control action. We develop an interface that converts predictive uncertainty into dynamics ambiguity sets that a controller can query. A residual Fourier neural operator supplies a one-step world model, and a perturbed-label replica supplies a plug-in spatial scale. A disjoint target-domain audit set then calibrates field-valued ellipsoids and simultaneous coordinate boxes without retraining either operator. Their exact support functions expose how two sets with matched field-level coverage can induce different robust actions. We embed these support functions in receding-horizon control for a persistently forced Burgers equation. Before fitting the learned controller, we test whether the benchmark requires active control. Across 100 independently sampled plants, finite-budget PDE-oracle MPC reduces mean cumulative cost from 0.5900 to 0.1610 and improves every matched case. This result validates the control task but does not establish superiority of a learned robust controller. The remaining multi-seed comparison is specified prospectively. The framework is offline-calibrated and online-applied; one-step conformal coverage is not presented as a closed-loop safety guarantee.

## 1 Introduction

Neural operators learn mappings between function spaces and amortize repeated PDE solution across input fields and parameters [14][5]. Fourier neural operators (FNOs) are particularly attractive as differentiable world models because their global spectral kernels support fast rollouts [16]. Prediction accuracy alone, however, does not determine the quality of a model-based control decision. A small field error can be costly when it aligns with a sensitive direction of the finite-horizon objective, whereas a larger orthogonal error can be nearly irrelevant [11][2].

Uncertainty quantification for neural operators has progressed through ensembles, probabilistic operators, approximate Bayesian models, and function-valued Gaussian approximations [22][6][19][20]. Conformal prediction provides a complementary calibration layer. Under exchangeability, split conformal prediction wraps a fixed score with a finite-sample marginal coverage guarantee [25][15][1]. Recent extensions address operator outputs and simultaneous fields [18][23][17]. Most closely related, Wang et al. use the disagreement between clean-label and perturbed-label FNOs to construct narrow simultaneous bands in a data-scarce Navier–Stokes setting [26]. We adopt that disagreement as a plug-in scale; it is not our architectural contribution.

Coverage specifies a containment event, not the consequence of that event for an action. Robust optimization instead asks how an uncertainty set interacts with an objective through its support function [4][13]. In control, conformal sets have been used for constraint tightening, reference generation, and adaptive model predictive control (MPC) [7][10][8]. These studies motivate a narrower question for PDE world models: once field uncertainty has been calibrated, which set geometry is effective for the control decision?

We address this gap by separating three objects that are often conflated: predictive coverage, ambiguity-set geometry, and closed-loop utility. The FNO is frozen before calibration. A small labelled target-domain audit split calibrates either a normalized ellipsoid or a simultaneous coordinate box. The controller then queries the same calibrated field uncertainty along finite-horizon adjoint sensitivities. This construction is *offline-calibrated and online-applied*; it does not update conformal quantiles during deployment.

The paper makes two methodological contributions and one evaluation-design contribution. First, it constructs state-action-dependent dynamics ambiguity sets from target-audited neural-operator residuals. Second, it derives exact support functions for the ellipsoid and box along decision-relevant adjoints, yielding a tractable robust MPC interface. Third, it specifies a paired, multi-seed evaluation that compares geometry at matched field-level coverage. The present results establish the prerequisite that the revised Burgers task is controllable and cannot be solved by passive viscous decay.

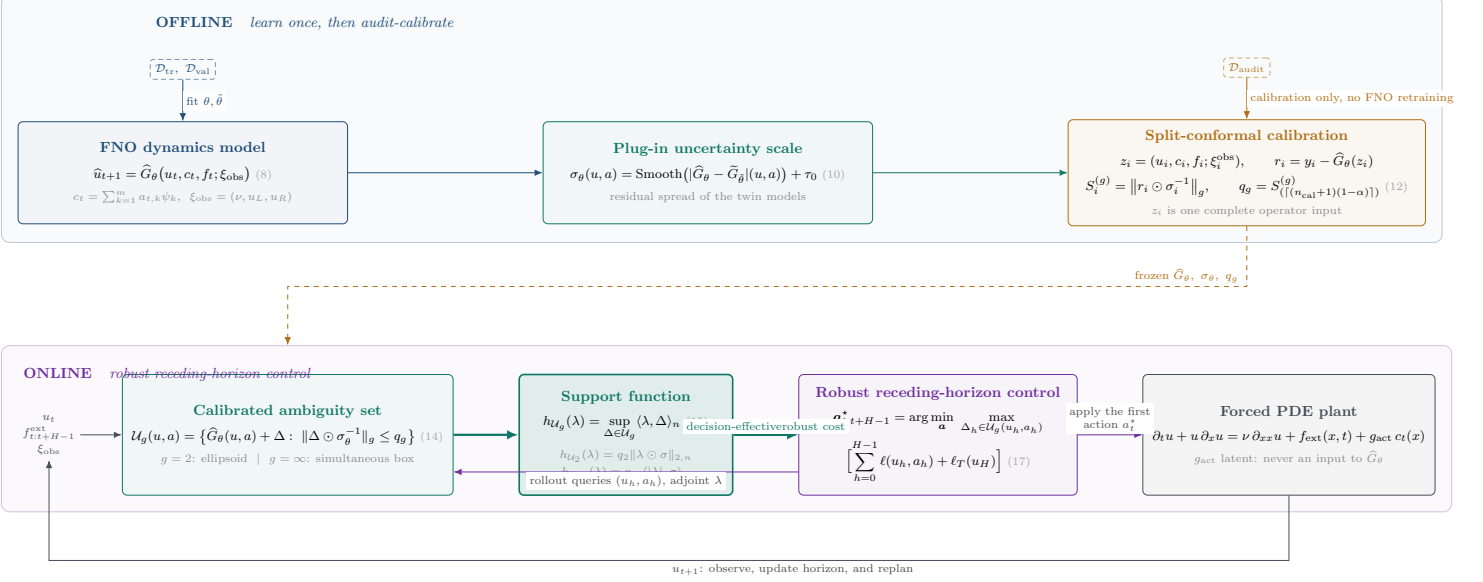


Figure 1: Overview of the proposed uncertainty-to-control framework. Offline, the FNO pair supplies a spatial scale and a disjoint target-domain audit split calibrates a field-valued dynamics set without FNO retraining. Online, robust receding-horizon control queries the support of the frozen set, applies the first action to the forced PDE plant, observes the next field, and replans. The bottom arrow is the control feedback loop; the dashed transfer denotes offline calibration rather than online conformal updating.

## 2 Problem formulation and preliminaries

### 2.1 Persistently forced controlled Burgers dynamics

We consider the viscous Burgers equation on  $x \in (0, 1)$ ,

$$\partial_t u + u \partial_x u = \nu \partial_{xx} u + f_{\text{ext}}(x, t) + g_{\text{act}} \sum_{k=1}^m a_{t,k} b_k(x), \quad u(0, t) = u_L, \quad u(1, t) = u_R. \quad (1)$$

The action is constant over a control interval  $\Delta T$ . We use two localized actuator profiles and a persistent forcing field,

$$b_k(x) \propto \exp\left[-\frac{(x - c_k)^2}{2s_b^2}\right], \quad f_{\text{ext}}(x, t) = A_f [\sin(\pi x) + 0.35 \sin(2\pi f_f t + \phi_f) \sin(2\pi x)]. \quad (2)$$

The endpoint values of each  $b_k$  and  $f_{\text{ext}}$  are set to zero on the numerical grid. The constants  $c_k$  and  $s_b$  are benchmark-design choices, not estimated parameters. Persistent forcing prevents the zero reference from being reached by viscous decay alone.

Let  $X_{u_L, u_R} = \{v \in H^1(0, 1) : v(0) = u_L, v(1) = u_R\}$  and let  $A = [-a_{\max}, a_{\max}]^m$ . The time- $\Delta T$  solution operator is

$$\mathcal{G}_\star : X_{u_L, u_R} \times A \times \Xi \longrightarrow X_{u_L, u_R}, \quad u_{t+1} = \mathcal{G}_\star(u_t, a_t; \xi), \quad (3)$$

where  $\xi$  contains observed quantities such as viscosity, boundaries, and the forcing forecast. The actuator gain  $g_{\text{act}}$  is latent to the learned model and creates a controlled dynamics shift. On a grid with  $n$  coordinates, sampling and interpolation give the learned finite-dimensional map

$$G_{\star, n} = P_n \mathcal{G}_\star I_n : \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^{d_\xi} \longrightarrow \mathbb{R}^n. \quad (4)$$

The finite-grid map is the object used in computation. We do not claim that a fixed-grid conformal result automatically lifts to the continuum.

For a horizon  $H$ , the controller minimizes

$$J(u_t, a_{t:t+H-1}) = \sum_{h=0}^{H-1} \left[ \frac{1}{n} \|u_{t+h} - u_{\text{ref}, t+h}\|_2^2 + \lambda_a \|a_{t+h}\|_2^2 \right] + \lambda_T \frac{1}{n} \|u_{t+H} - u_{\text{ref}, t+H}\|_2^2. \quad (5)$$

Only the first optimized action is applied before the state is observed again, as in standard receding-horizon MPC [21].

## 2.2 Split conformal prediction on a field

Split conformal prediction separates model fitting from calibration. Let  $\mathcal{D}_{\text{cal}} = \{(z_i, y_i)\}_{i=1}^{n_{\text{cal}}}$  be disjoint from training and let  $S(z, y)$  be a scalar nonconformity score for one complete output field:

$$S_i = S(z_i, y_i). \quad (6)$$

For target miscoverage  $\alpha$ , define the finite-sample quantile

$$q_\alpha = S_{(k)}, \quad k = \lceil (n_{\text{cal}} + 1)(1 - \alpha) \rceil, \quad (7)$$

with the usual atom-at-infinity convention when  $k > n_{\text{cal}}$ .

**Proposition 1** (Fixed-grid split-conformal coverage). *If the calibration pairs and one test pair are exchangeable, and the fitted predictor and score rule are frozen before calibration, then the prediction set  $C_\alpha(z) = \{y : S(z, y) \leq q_\alpha\}$  satisfies  $\Pr\{Y_{n_{\text{cal}}+1} \in C_\alpha(Z_{n_{\text{cal}}+1})\} \geq 1 - \alpha$ .*

The result follows from the exchangeable rank of the test score among the  $n_{\text{cal}} + 1$  scores [25][15]. It is marginal over a new field sampled from the calibration population. Source calibration does not retain this guarantee after arbitrary target-domain shift. We therefore distinguish a source calibration split from a labelled target-domain audit split. Neither guarantee implies simultaneous validity across an adaptive MPC trajectory [3][24].

## 3 Decision-effective conformal ambiguity sets

### 3.1 Residual FNO world model and uncertainty scale

The one-step world model is a residual FNO

$$\hat{u}_{t+1} = \hat{G}_\theta(u_t, a_t; \xi_{\text{obs}}) = u_t + \delta_\theta(u_t, c_t, f_t; \xi_{\text{obs}}), \quad c_t(x) = \sum_{k=1}^m a_{t,k} b_k(x). \quad (8)$$

The input channels contain the current state, commanded control field, forcing field, observed physical parameters, and coordinate. Each residual block combines a truncated Fourier convolution with a pointwise linear path and an identity skip. The FNO is a differentiable surrogate, not the proposed architectural novelty [16][12].

A replica  $\tilde{G}_{\tilde{\theta}}$  uses the same proper-training inputs and architecture but one fixed Gaussian-perturbed copy of the targets:

$$\tilde{\theta} = \arg \min_{\tilde{\theta}} \sum_{(z,y) \in \mathcal{D}_{\text{tr}}} \|\tilde{G}_{\tilde{\theta}}(z) - (y + \varepsilon)\|_2^2, \quad \varepsilon \sim \mathcal{N}(0, \sigma_{\text{pert}}^2 I). \quad (9)$$

Following the perturbation construction of Wang et al. [26], the smoothed disagreement supplies a plug-in scale

$$\sigma_\theta(z) = \text{Smooth}\left(|\hat{G}_\theta(z) - \tilde{G}_{\tilde{\theta}}(z)|\right) + \tau_0, \quad (10)$$

where the positive floor  $\tau_0$  and smoothing rule are fixed from proper training predictions. The scale can improve efficiency when disagreement localizes residuals, but calibration does not require it to be a consistent conditional error estimator.

### 3.2 Target-audited field geometries

After freezing both models, define the residual  $r_i = y_i - \hat{G}_\theta(z_i)$  on the target audit split. For  $g \in \{2, \infty\}$ , use one scalar score per complete field,

$$S_i^{(g)} = \|r_i \odot \sigma_\theta(z_i)^{-1}\|_{g,n}, \quad \|v\|_{2,n} = \sqrt{n^{-1} \sum_j v_j^2}, \quad \|v\|_{\infty,n} = \max_j |v_j|. \quad (11)$$

The target-audit multiplier is

$$q_g = S_{(\lceil (n_{\text{audit}} + 1)(1 - \alpha) \rceil)}^{(g)}. \quad (12)$$

For reference, the induced prediction set is

$$C_g(z) = \left\{ y : \left\| (y - \hat{G}_\theta(z)) \odot \sigma_\theta(z)^{-1} \right\|_{g,n} \leq q_g \right\}. \quad (13)$$

Equivalently, the controller receives the state-action dynamics ambiguity set

$$\mathcal{U}_g(u, a) = \left\{ \hat{G}_\theta(u, a) + \Delta : \left\| \Delta \odot \sigma_\theta(u, a)^{-1} \right\|_{g,n} \leq q_g \right\}. \quad (14)$$

The  $g = 2$  set is an anisotropic ellipsoid. The  $g = \infty$  set is a simultaneous coordinate box. An audit- $L^2$  baseline sets  $\sigma_\theta \equiv 1$  and therefore discards spatial localization.

The ellipsoid and box can be tuned to the same empirical field-membership coverage, but they do not encode the same event. The box controls every coordinate within a field under a max score. The ellipsoid controls joint membership in a normalized  $L^2$  region. We therefore use the precise phrase *matched field-level coverage*, rather than calling both guarantees simultaneous coordinate coverage.

### 3.3 Support functions and robust MPC

Let  $\lambda \in \mathbb{R}^n$  be the sensitivity of the finite-horizon cost to a one-step state perturbation and define  $\langle v, w \rangle_n = n^{-1} v^\top w$ . The support function of the residual part of Equation (14) is

$$h_{\mathcal{U}_g}(\lambda) = \sup_{\|\Delta \odot \sigma^{-1}\|_{g,n} \leq q_g} \langle \lambda, \Delta \rangle_n = \begin{cases} q_2 \|\lambda \odot \sigma\|_{2,n}, & g = 2, \\ q_\infty \|\lambda \odot \sigma\|_{1,n}, & g = \infty, \end{cases} \quad (15)$$

where  $\|v\|_{1,n} = n^{-1} \sum_j |v_j|$ . These formulas follow from dual norms and are exact for the local linear functional  $\langle \lambda, \Delta \rangle_n$  [4].

For a nominal rollout  $\hat{u}_{h+1} = \hat{G}_\theta(\hat{u}_h, a_h)$ , reverse-mode differentiation gives the discrete adjoint recursion

$$\lambda_H = \nabla_u \ell_T(\hat{u}_H), \quad \lambda_h = \nabla_u \ell(\hat{u}_h, a_h) + \left[ D_u \hat{G}_\theta(\hat{u}_h, a_h) \right]^\top \lambda_{h+1}. \quad (16)$$

The first-order robust planning problem is

$$a_{t:t+H-1}^* = \arg \min_{a_{t:t+H-1}} \left[ J(\hat{u}_t, a_{t:t+H-1}) + \sum_{h=0}^{H-1} h_{\mathcal{U}_g}(\lambda_{h+1}) \right]. \quad (17)$$

The support term is decision-effective because it weights uncertainty by the objective sensitivity. It is not a certified solution of the nonlinear worst-case FNO rollout. State-dependent scales and repeated FNO composition make that inner problem high-dimensional and nonconvex. The present paper therefore studies the transparent adjoint approximation and does not use projected-gradient search as evidence of a global robust optimum.

### 3.4 Practical implementation

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**Algorithm 1** Offline calibration and online robust receding-horizon control

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**Require:** Proper-training data  $\mathcal{D}_{\text{tr}}$ , validation data  $\mathcal{D}_{\text{val}}$ , target audit data  $\mathcal{D}_{\text{audit}}$ , level  $\alpha$

- 1: Train  $\hat{G}_\theta$  on clean targets and  $\tilde{G}_\theta$  on perturbed targets; select checkpoints using only  $\mathcal{D}_{\text{val}}$ .
  - 2: Freeze both operators. Fix the smoothing rule and  $\tau_0$  from proper-training predictions.
  - 3: Compute  $\sigma_\theta$ , residuals, and scores on  $\mathcal{D}_{\text{audit}}$  using Equations (10) and (11).
  - 4: Compute  $q_2$  and  $q_\infty$  using Equation (12); store weights, scale settings, split metadata, and random seeds.
  - 5: **for** each online control time  $t$  **do**
  - 6:   Observe  $u_t$  and the available forcing and physical parameters.
  - 7:   Optimize Equation (17) with the frozen model, scale, and calibration multiplier.
  - 8:   Apply only the first action, observe  $u_{t+1}$ , shift the horizon, and replan.
  - 9: **end for**
  - 10: Keep all final coverage and control-test labels untouched.
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Calibration is offline and uses labelled target-domain transitions. The online loop updates the state and action horizon, not the FNO weights or conformal quantiles. This distinction separates the method from adaptive or sliding-window conformal control [10][8].

## 4 Experimental design and task validation

### 4.1 Numerical benchmark and test population

Equation (1) is solved on 48 grid points. The control interval is 0.025, and each interval uses finite-difference steps of  $5 \times 10^{-4}$  with an upwind convective flux and centered diffusion. The two Gaussian actuators have centers 0.28 and 0.72 and width 0.10. Actions lie in  $[-2, 2]^2$ . The reference is zero,  $\lambda_a = 0.002$ , and  $\lambda_T = 4$ . These are fixed simulator and objective choices rather than estimated quantities.

The task-validity test draws 100 independent plants from the joint distribution in Table 1. Within each draw, uncontrolled dynamics and PDE-oracle MPC share the initial field and all physical parameters. The term *PDE-oracle* denotes direct numerical-model access, not a globally optimal policy. Its cross-entropy-method optimizer uses horizon six, 64 candidates, eight elites, and three iterations [9]. Each controller is evaluated for 20 control steps, or a physical horizon of 0.5.

Table 1: Independent joint test distribution for the persistent-forcing Burgers benchmark.

| Quantity                              | Distribution       |
|---------------------------------------|--------------------|
| Viscosity $\nu$                       | Unif(0.008, 0.018) |
| Each boundary value                   | Unif(−0.03, 0.03)  |
| Latent actuator gain $g_{\text{act}}$ | Unif(0.65, 1.35)   |
| Forcing amplitude $A_f$               | Unif(0.40, 0.80)   |
| Forcing frequency $f_f$               | Unif(0.50, 1.50)   |
| Forcing phase $\phi_f$                | Unif(0, $2\pi$ )   |
| Initial-field amplitude               | Unif(0.03, 0.12)   |

### 4.2 Persistent forcing creates a meaningful control task

Table 2: Task-validity results on 100 matched, independently sampled plants. Intervals use 5,000 nonparametric bootstrap resamples. Lower cost is better.

| Controller     | Mean cost (95% CI)      | Median | $p90$ cost (95% CI)     |
|----------------|-------------------------|--------|-------------------------|
| Uncontrolled   | 0.5900 (0.5176, 0.6661) | 0.5249 | 1.0599 (0.9632, 1.3942) |
| PDE-oracle MPC | 0.1610 (0.1410, 0.1812) | 0.1377 | 0.2875 (0.2322, 0.3401) |

PDE-oracle MPC reduced cost in every matched case (Figure 2). The paired mean difference, oracle minus uncontrolled, was  $-0.4289$ , with a 95% paired-bootstrap interval of  $(-0.4883, -0.3702)$ . The empirical  $p90$  difference was  $-0.7724$ , with a 95% bootstrap interval of  $(-1.0610, -0.7166)$ . These results show that the revised task rewards active control rather than passive decay.

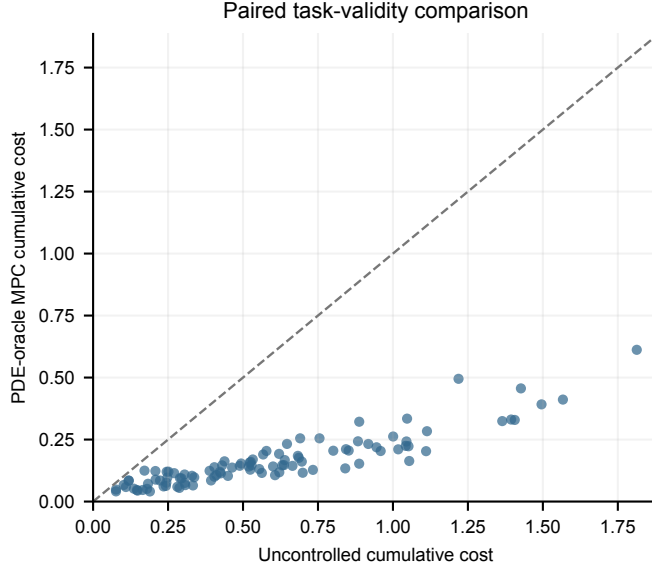


Figure 2: Paired cumulative costs for 100 independently sampled plants. Every point lies below the identity line, indicating lower cost for finite-budget PDE-oracle MPC than for zero control on the same plant.

A prespecified higher-budget CEM audit on the first 20 cases used horizon eight, 128 candidates, 16 elites, and five iterations. It reduced mean oracle cost from 0.1662 to 0.1547 and improved all 20 paired cases. The main conclusion therefore did not arise from the smaller CEM budget. This check still does not prove global optimality.

### 4.3 Prespecified evaluation required for the full claim

The task-validity result is necessary but not sufficient. The full empirical claim requires five independently trained FNO seeds and 100–200 independent joint-shift trajectories per seed. All controllers will use identical test trajectories and common CEM random numbers. The comparison is restricted to uncontrolled dynamics, PDE-oracle MPC, nominal FNO MPC, audit- $L^2$  MPC, ellipsoid-adjoint MPC, and box-adjoint MPC.

Source calibration will be compared with target audit calibration without updating FNO weights. Audit sizes  $n_{\text{audit}} \in \{20, 50, 100, 200, 300\}$  will be resampled 10–20 times. Scale ablations will compare a constant scale, twin-FNO disagreement, a deep ensemble, and a residual-quantile scale head under transparent label budgets. The primary control statistics are mean, median, and  $p_{90}$  cumulative cost, paired differences from nominal MPC, and average control effort. Coverage, set width, adjoint support, and control cost will be reported together.

The central test is deliberately narrow: at matched field-level coverage, do the ellipsoid and box produce different adjoint support, actions, and upper-tail closed-loop cost? This claim will not be made until the prespecified multi-seed experiment is complete.

## 5 Limitations and future scope

The current evidence validates the benchmark, not the learned robust controller. Completing the prespecified five-seed study is therefore the immediate priority. Nominal FNO MPC must first achieve a sensible cost between the PDE-oracle and uncontrolled baselines. Only then can differences among calibrated ambiguity geometries be interpreted as evidence about robust control rather than inadequate model learning.

The target-domain audit uses offline labelled transitions. Future work should study sequential recalibration under nonstationary drift, while accounting for the dependence created by feedback. Weighted or adaptive conformal methods are relevant here, but their assumptions and guarantees differ from the split protocol used in this paper [3][10].

The present ambiguity model is local and stagewise. Operator errors can be temporally correlated, and a controller can visit states that are absent from the audit distribution. A useful extension is a trajectory-coupled ambiguity set or an online residual model that represents cross-time dependence. Any result of this type must distinguish empirical rollout coverage from a genuine closed-loop guarantee.

The adjoint support is exact only for the local linear objective perturbation. Nonlinear FNO rollouts remain high-dimensional and nonconvex. Certified curvature control, convex outer bounds, or verified relaxations would be needed before describing the robust inner problem as globally solved.

Finally, the current controlled benchmark is one-dimensional. A controlled two-dimensional flow benchmark would test whether the same set geometry remains useful at larger state dimension. FNO, Toeplitz neural operators, and other PDE surrogates can then be compared while keeping calibration and the robust-control interface fixed. Such comparisons would test architectural portability rather than redefine the contribution.

## 6 Conclusion

This work isolates the missing interface between neural-operator uncertainty and robust PDE control. A frozen FNO pair supplies a spatial scale, a disjoint target audit calibrates a field-valued dynamics set, and exact support functions connect its geometry to finite-horizon adjoint sensitivities. The persistently forced Burgers benchmark passes the prerequisite control-value test: finite-budget PDE-oracle MPC outperformed zero control on all 100 matched plants. The evidence does not yet establish that a calibrated learned controller improves closed-loop performance. That conclusion depends on the prespecified multi-seed, matched-coverage comparison. The intended contribution is therefore precise: predictive coverage becomes decision-relevant only through the geometry that the controller actually queries.

## Data availability

All Burgers samples are generated by the numerical solver in the repository. The task-validation case-level data, summary statistics, and figure source files are included with the code. No external dataset is used for the results reported in the main text.

## Code availability

The simulator, task-validity experiment, calibration components, controller implementations, audit script, and saved results are available at <https://github.com/YaowenKANG666/decision-calibrated-pde-control>.

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